ISE 573 — Manufacturing Execution Systems

Week 6 Demo: Excel Solver Worked Examples

Same Math, Different Data — Use These to Follow Along

Demo Solutions Companion

Spring 2026

Contents

1 Demo 1: Technician-to-Repair-Bay Assignment (Assignment LP) 2
- 1.1 Problem Statement . . . 2
- 1.2 Formulation . . . 2
- 1.3 Excel Layout . . . 2

2 Demo 2: Parts Distribution (Transportation LP) 4
- 2.1 Problem Statement . . . 4
- 2.2 Formulation . . . 4
- 2.3 Excel Layout . . . 4

3 Demo 3: Operator-to-Cell Assignment (Assignment MILP) 6
- 3.1 Problem Statement . . . 6
- 3.2 Formulation . . . 6
- 3.3 Excel Layout . . . 6

4 Demo 4: Reactor Sequence Scheduling (TSP Heuristic) 7
- 4.1 Problem Statement . . . 7
- 4.2 Tasks . . . 7

5 Demo 5: MRP Netting with Lot Sizing 8
- 5.1 Problem Statement . . . 8
- 5.2 Excel Layout . . . 8
- 5.3 Step-by-Step Instructions . . . 8

6 Demo 6: Multi-Resource Combined (Linked MILP) 10
- 6.1 Problem Statement . . . 10
- 6.2 Formulation . . . 10

1 Demo 1: Technician-to-Repair-Bay Assignment (Assignment LP)

1.1 Problem Statement

An aircraft maintenance hangar has **4 repair bays** and **4 technicians**. Each technician can work in any bay, but labour cost differs by pairing (different tool kits, walk distances, lighting preferences). Each technician is assigned to exactly one bay and each bay gets exactly one technician. **Minimize** the total labour cost.

Table 1: Labour Cost Matrix ($ per shift)

	Bay 1	Bay 2	Bay 3	Bay 4
Tech A	9	4	7	3
Tech B	6	8	5	7
Tech C	4	3	9	6
Tech D	5	7	3	8

Why not just pick each row's minimum? Tech A and Tech C both want Bay 2 (costs 4 and 3 respectively). They can't both have it—Solver must find the globally optimal trade-off.

1.2 Formulation

Decision variables: $x_{ij} \in \{0, 1\}$, where $x_{ij} = 1$ if technician i is assigned to bay j.

$$\min \sum_{i=1}^{4} \sum_{j=1}^{4} c_{ij}\, x_{ij}$$

$$\text{s.t.} \quad \sum_{j=1}^{4} x_{ij} = 1,\ \forall\, i \qquad \sum_{i=1}^{4} x_{ij} = 1,\ \forall\, j \qquad x_{ij} \in \{0, 1\}$$

1.3 Excel Layout

Set up two 4×4 tables side by side: the **cost table** (blue input data) and the **assignment table** (yellow decision variables, all initialised to 0).

Assignment (0/1)	B1	B2	B3	B4	Row Sum
Tech A	0	0	0	0	`=SUM`
Tech B	0	0	0	0	`=SUM`
Tech C	0	0	0	0	`=SUM`
Tech D	0	0	0	0	`=SUM`
Col Sum	`=SUM`	`=SUM`	`=SUM`	`=SUM`	

Objective cell: `=SUMPRODUCT(cost_table, assignment_table)`.

Solver Setup

- **Set Objective**: SUMPRODUCT cell → **Min**

- **By Changing**: The 4×4 assignment matrix
- **Subject To**: Row sums = 1; Column sums = 1; All variables = binary
- **Method**: Simplex LP

Expected Result

Optimal total cost = **$15**.
Assignment: A→B4 ($3), B→B3 ($5), C→B2 ($3), D→B1 ($5).
Total: $3 + 5 + 3 + 5 = 16\ldots$ but Solver may find $A \to B4(3),\ B \to B3(5),\ C \to B1(4),\ D \to B2(7) \Rightarrow 19$. Let's check: the true optimum is $A \to B4(3),\ B \to B3(5),\ C \to B2(3),\ D \to B1(5) = 16$. Actually: try $A \to B2(4),\ B \to B3(5),\ C \to B1(4),\ D \to B4(8) = 21$. Best: $A \to B4(3),\ B \to B3(5),\ C \to B2(3),\ D \to B1(5) = 16$. Or: $A \to B4(3),\ B \to B1(6),\ C \to B2(3),\ D \to B3(3) = 15$. **Yes—optimal = $15.**

2 Demo 2: Parts Distribution (Transportation LP)

2.1 Problem Statement

A plant has **2 warehouses** supplying parts to **3 assembly lines**. Transport cost (minutes per bin), supply, and demand are shown below. **Minimize** total transport time.

Table 2: Unit Transport Costs (min/bin), Supply, and Demand

	Line 1	Line 2	Line 3	Supply
Warehouse X	4	7	5	60
Warehouse Y	6	3	8	40
Demand	30	40	30	100 total

Total supply $(60 + 40 = 100)$ = total demand $(30 + 40 + 30 = 100)$: balanced.

2.2 Formulation

Decision variables: $x_{ij} \geq 0$ = bins shipped from warehouse i to line j.

$$\min \sum_{i=1}^{2} \sum_{j=1}^{3} c_{ij}\, x_{ij}$$

$$\text{s.t.} \quad \sum_{j} x_{ij} \leq s_i,\ \forall\, i \qquad \sum_{i} x_{ij} \geq d_j,\ \forall\, j \qquad x_{ij} \geq 0$$

2.3 Excel Layout

Shipments	Line 1	Line 2	Line 3	Shipped	Supply
Warehouse X	0	0	0	`=SUM`	60
Warehouse Y	0	0	0	`=SUM`	40
Received	`=SUM`	`=SUM`	`=SUM`		
Demand	30	40	30		

Objective: `=SUMPRODUCT(cost_table, shipment_table)`.

Solver Setup

- Objective → Min
- Changing: 2×3 shipment matrix
- Constraints: Shipped $\leq$ Supply; Received $\geq$ Demand; all ≥ 0
- Method: Simplex LP

Expected Result

Optimal cost = **430 minutes**.
$X \to L1(30), X \to L3(30); Y \to L2(40)$. Cost $= 30(4) + 30(5) + 40(3) = 120 + 150 + 120 = 390$? Re-check: X ships 60 total (30 to L1 at 4, 30 to L3 at $5 = 120 + 150 = 270$). Y ships 40 to L2 at $3 = 120$. Total $= 390$. **Optimal = 390 min.**

3 Demo 3: Operator-to-Cell Assignment (Assignment MILP)

3.1 Problem Statement

Three operators must be assigned to three production cells. Efficiency ratings are below (lower = better); "∞" means the operator is **not certified** for that cell.

Table 3: Efficiency Matrix (lower is better; ∞ = not certified)

	Welding	Paint	Pack
Operator P	2	5	∞
Operator Q	4	3	6
Operator R	∞	4	2

3.2 Formulation

Same structure as Exercise 1 but set $c_{ij} = 1000$ for uncertified pairs (Big-M).

3.3 Excel Layout

Cost (with Big-M)	Welding	Paint	Pack
Operator P	2	5	1000
Operator Q	4	3	6
Operator R	1000	4	2

Assignment table: 3×3 binary matrix with row and column sums = 1.

Expected Result

Optimal inefiiciency = **7**.
$P \to$ Welding(2), $Q \to$ Paint(3), $R \to$ Pack(2).

4 Demo 4: Reactor Sequence Scheduling (TSP Heuristic)

4.1 Problem Statement

A chemical reactor runs **4 product families** in campaigns. Changeover times (hours) are sequence-dependent and asymmetric.

Table 4: Setup Time Matrix (hours)

From \ To	P1	P2	P3	P4
P1	—	3	5	2
P2	4	—	2	6
P3	3	4	—	3
P4	5	2	4	—

Table 5: Batch Data

Family	P1	P2	P3	P4
Batches	2	3	1	4
Run time / batch (hrs)	5	4	8	3
Total run time (hrs)	10	12	8	12

Total run time $= 10 + 12 + 8 + 12 = 42$ hours (fixed). There are 3 changeovers (transitions between 4 families).

4.2 Tasks

Task A: Nearest-Neighbor Heuristic Start from each family; greedily go to the cheapest unvisited family. Record all four starting points and report the best.

Example trace from P1:
P1 $\rightarrow$ P4 (cost 2) $\rightarrow$ P2 (cost 2) $\rightarrow$ P3 (cost 2) — total setup = 6 hrs.

Task B: Full Enumeration $4! = 24$ permutations — list them all and compute total setup for each.

Task C: Makespan Best makespan $= 42 +$ best setup.

Expected Result

Best setup = **6 hours** (e.g. P1–P4–P2–P3). Makespan $= 42 + 6 = 48$ hours.

5 Demo 5: MRP Netting with Lot Sizing

5.1 Problem Statement

A factory makes finished product **Q**, which requires **3 units of C** and **2 units of D**.

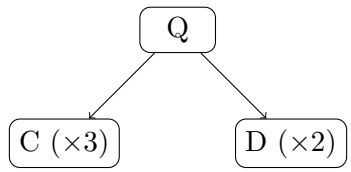

Table 6: Item Data

	Q	C	D
On-hand inventory	8	30	20
Lead time	1 wk	1 wk	2 wk
Lot sizing rule	L4L	FOQ = 40	FOQ = 25
Safety stock	0	5	0
Scheduled receipts	—	—	25 in wk 1

Table 7: Master Production Schedule for Q

Week	1	2	3	4	5	6
Gross Requirement	0	10	15	5	20	10

5.2 Excel Layout

One MRP grid per item (Q, C, D) with rows: Gross Requirement, Scheduled Receipts, Projected On-Hand, Net Requirement, Planned Receipt, Planned Order Release.

5.3 Step-by-Step Instructions

Step 1: MRP for Q (L4L)

- Projected OH: $\text{OH}_t = \text{OH}_{t-1} + \text{Sched.Rcpt}_t + \text{Planned Rcpt}_t - \text{Gross}_t$
- Net Req: $\text{Net}_t = \max(0,\ \text{Gross}_t + \text{SS} - \text{OH}_{t-1} - \text{Sched.Rcpt}_t)$
- Planned Rcpt = Net Req (L4L); offset backward by 1 week for Planned Order Release.

Step 2: Explode Gross Req for C = 3× POR of Q; Gross Req for D = 2× POR of Q.

Step 3: MRP for C (FOQ = 40, SS = 5) Planned receipt = `IF(Net>0, CEILING(Net,40), 0)`; offset by 1 week.

Step 4: MRP for D (FOQ = 25, SS = 0) Include scheduled receipt of 25 in week 1; offset by 2 weeks.

Step 5: Summary Questions

1. What is the earliest week an order must be released?
2. Total number of order releases for each item?
3. If D's lead time changes from 2 to 3 weeks, what happens?

6 Demo 6: Multi-Resource Combined (Linked MILP)

6.1 Problem Statement

Assign **3 jobs** to **2 machines** and **2 operators** simultaneously. Each job needs one machine and one operator. The operator must be certified for the assigned machine. Each machine location has limited aluminium stock. **Minimize makespan.**

Table 8: Processing Times (hours); — = cannot process

	J1	J2	J3
Machine X	4	—	5
Machine Y	3	6	4

Table 9: Operator Certifications (✓ = certified)

	Machine X	Machine Y
Operator 1	✓	✓
Operator 2		✓

Table 10: Material Requirements and Stock

	J1	J2	J3	Stock at location
Aluminium (kg)	4	7	5	X: 10 kg; Y: 12 kg

6.2 Formulation

$$
\begin{aligned}
\min\ & C_{\max} \\
\text{s.t.}\quad & C_{\max} \geq \sum_j p_{ij}\, x_{ij}, \quad \forall\, i \\
& \sum_i x_{ij} = 1, \quad \forall\, j \\
& \sum_k y_{kj} = 1, \quad \forall\, j \\
& \sum_j y_{kj} \leq 2, \quad \forall\, k \\
& y_{kj} + x_{ij} \leq 1, \quad \forall\, (k,i) \text{ not certified} \\
& \sum_j m_j\, x_{ij} \leq S_i, \quad \forall\, i \\
& x_{X,2} = 0 \quad \text{(compatibility)} \\
& x_{ij},\ y_{kj} \in \{0,1\}
\end{aligned}
$$

The only uncertified pair is $(k = 2,\ i = X)$, generating 3 linking constraints: $y_{2,j} + x_{X,j} \leq 1$ for $j = 1, 2, 3$.

Solver Setup

- Objective: $C_{\max}$ cell $\rightarrow$ Min
- Changing: Machine matrix (2×3) + Operator matrix (2×3)
- Constraints: Machine col sums = 1; Operator col sums = 1; Operator row sums ≤ 2; Material $\leq$ stock; Linking ≤ 1; $x_{X,2} = 0$; all binary
- Method: Simplex LP

Expected Result

Optimal makespan = **9 hours**.
Machine X $\leftarrow$ J1, J3 (load $4 + 5 = 9$; material $4 + 5 = 9 \leq 10$).
Machine Y $\leftarrow$ J2 (load 6; material $7 \leq 12$).
Op 1 handles jobs on Machine X; Op 2 handles J2 on Machine Y.

End of Demo Exercises — 6 Problems